

Not Found

The requested URL /page.html was not found on this server.

Apache/2.2.3 (Unix) mod_ssl/2.2.3 OpenSSL/0.9.7g DAV/2 PHP/5.1.2
Server at localhost Port 80

This error message can be generated by requesting a non-existent URL. After the common message that shows a page as not found, there is information about the Web server version, OS, modules and other products used. This information can be very useful to attackers.

Other methods include guessing the files with no direct link pointing to them. The best example of this is guessing the URL for the admin interface of the Web application—for example, guessing the URL www.example.com/wp-admin is quite easy.

Time for security

1. Disable WebDAV in Apache if you don't need it. To do this, find the following entries in *httpd.conf* and change 'On' to 'Off'.

```
<IfDefine DAV>
DAV On
</IfDefine>
```

By default, */usr/local/httpd/htdocs* is the only directory with the *IfDefine DAV* directive. Other directories that have this directive will also need to be changed. After this, restart Apache.

2. When dealing with information disclosures in HTML comment tags, it would not be appropriate to deny the entire request for a Web page due to comment tags. To handle this, there's a really cool feature in Apache 2.0, called filters (see http://httpd.apache.org/docs-2.0/mod/mod_ext_filter.html). The basic premise of filters is that they read from standard input and print to standard output. This feature becomes intriguing from a security perspective when dealing with this type of information disclosure prevention.
3. The application should never return verbose error messages or debug information to the user's browser. When an unexpected event occurs (such as an error in a database query, a failure to read a file from disk, or an exception in an external API call), the application should return the same generic message, informing the user that an error occurred.
4. In Apache, custom error pages can be configured using the *ErrorDocument* directive in *httpd.conf*. For example: *ErrorDocument 500 /generalerror.html*. This will redirect users to *generalerror.html*, which can be your own modified error page.
5. Disable or limit detailed error handling. In particular, do not display debug information, stack traces, or path information to end-users. Moreover, overriding the default error handler so that it always returns '200'

(OK) error screens reduces the ability of automated scanning tools to determine if a serious error occurred.

6. Lastly, programmers need to be more aware about whether they are unintentionally providing any crucial information.

We will deal with other dangerous attacks on Web applications and how to secure Apache in the next article. Always remember: know hacking, but no hacking. **END** 

Further reading

- [1] http://www.amanet.org/training/media_files/2010/the-secrets-of-search/Google-Cheat-Sheet.pdf
- [2] <http://netsecurity.about.com/b/2007/01/15/information-leakage-from-google-hacking.htm>

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